DHANANJAYA SAMANTA SINGHAR

Software Engineer III

9+ Years | Senior Backend Engineer | Distributed Systems • Microservices • Event-Driven Architecture • System Design | Java 21 • Spring Boot • Kafka • AWS • Docker • Kubernetes • CI/CD • Security • Observability

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PROFESSIONAL SUMMARY

Experienced Backend, Platform & Data Engineering Professional with **9 years** of experience, **building scalable distributed systems, cloud-native applications, and event-driven platforms**. Expertise in Java, Spring Boot, WebFlux, Kafka, AWS, Docker, Kubernetes, Python, Spark, and Snowflake, with strong ownership of backend architecture, platform engineering, data pipelines, and cloud-native solutions. Proven experience in system design, performance optimization, scalability, resiliency, observability, and enterprise data platforms, with exposure to **GenAI, LLMs, RAG**, **LangChain, LangGraph**, AI Agents, and MCP. Currently working as **AVP / Software Engineer III at JPMorgan Chase & Co**., contributing to Asset & Wealth Management (AWM) Technology – **Data Engineering & Analytics.**

TECHNICAL EXPERTISE

- **Backend & Platform Engineering -** Experienced in building scalable distributed systems, **microservices**, and cloud-native applications using **Java 21, Spring Boot, WebFlux, Kafka**, **AWS, Docker, and Kubernetes**, with a focus on performance, reliability, and scalability.
- **System Design & Architecture -** Strong experience in end-to-end **system design**, API architecture, **event-driven systems**, **asynchronous processing, transaction, resiliency, observability.**
- **Microservices Design Patterns:** Saga (**Temporal Workflow Orchestration**), **Circuit Breaker**, Retry, Distributed Tracing, Dead Letter Queue (DLQ/DLT), **CQRS, Transactional Outbox**.
- **API & Integration Engineering -** Experienced in designing and developing **REST APIs**, **GraphQL APIs**, distributed integrations, and **event-driven** communication for enterprise applications.
- **Security & Application Engineering -** Hands-on experience with **OAuth2**, **JWT**, Spring **Security**, HTTPS, SSL/TLS, and secure microservice communication.
- **Performance & Reliability Engineering -** Experience with **performance tuning**, **concurrency**, fault tolerance, retry mechanisms, **asynchronous processing**, **scalability**, and high-availability systems.
- **Observability & Production Engineering -** Experience with **logging, monitoring, metrics, tracing**, and production troubleshooting using **Grafana**, and **CloudWatch**.
- **Cloud & Platform Engineering -** Experienced in developing and **modernizing cloud-native platforms** using **AWS**, **Kubernetes**, **Docker**, and **CI/CD**, with a focus on automation and operational reliability.
- **Data Engineering & Distributed Data Systems** – Experience building data-intensive backend services, ETL pipelines, and distributed data platforms using Apache **Spark, Snowflake,** Apache **Iceberg,** Parquet, **AWS S3**.
- **GenAI, NLP & AI-Driven Engineering:** Building intelligent AI solutions using **LLMs, RAG, AI Agents, LangChain, LangGraph, MCP, and Agentic AI**, leveraging text preprocessing, tokenization, offline **TF-IDF-based vectorization**, cosine similarity, and **MongoDB for vector storage and retrieval** to enable document similarity analysis, relevance ranking, automation, and developer productivity
- **Technical Leadership & Project Ownership -** Demonstrated ability to handle requirements, stakeholder communication, **architecture decisions**, and end-to-end project delivery, **taking ownership** from solution design through production.

WORK EXPERIENCE

- Currently working in **J.P Morgan Chase & Co**, Bangalore, India as Software Engineer III from Jul-2024.
- Formerly worked in **A.P. Moller MAERSK** as a Software Engineer JL3 from Oct-2021 to Jul-2024.
- Formerly worked in **Lowe's India** as a Software Engineer from Mar-2020 to Oct-2021.
- Formerly worked in **ITC Infotech** as an Associate IT Consultant from Jun-2019 to Mar-2020.
- Formerly worked in **Allianz Technology SE** as a Software Engineer from Sep-2017 to May-2019

PROJECT DETAILS

— J.P Morgan Chase & Co. *Software Engineer III*

07/2024 – Present *Bangalore, India*

Key Contributions:

- Designed and developed scalable **Java 21** and **Spring Boot microservices**, exposing **REST** and **GraphQL** APIs for enterprise Asset & Wealth Management platforms.
- Built **high-throughput, event-driven processing** pipelines using Apache Kafka, implementing asynchronous communication, concurrency optimization and fault-tolerant processing.
- Improved event-processing **throughput by 10x,** scaling from approximately 200K to **2-3M events per hour** using Java 21 **Virtual Threads, JVM tuning** and **concurrency optimization.**
- Worked with **Drools Rule** to manage **business rules** and support **rule-driven application workflows**.
- Worked extensively on high-volume **transaction processing,** handling millions of records per hour with a focus on performance, reliability, **retry mechanism** and scalability.
- Designed and developed snapshot data platform supporting **dynamic ingestion** of **SQL(Postgres)** and **NoSQL(MongoDB)** data sources and **large-scale data processing**.
- Developed distributed **ETL pipelines** using Python, **PySpark** and AWS **EMR** to process large datasets and publish curated data into **Snowflake**.
- Performed **data validation, reconciliation**, **troubleshooting**, and production support across data-processing workflows to ensure data quality, consistency, reliability, and system stability.
- Took technical ownership of backend platform components, contributing to **architecture** and **platform design** decisions for scalable and resilient distributed systems.
- Collaborated with **data owners**, **business stakeholders**, and internal platform teams to understand data requirements, resolve data issues, and deliver reliable data solutions.
- Contributed to **production deployments**, **observability**, troubleshooting, **performance optimization** and **technical architecture decisions.**
- Developed GenAI-powered automation workflows and AI agents using Python, **LLMs, RAG, LangChain** and **LangGraph** for automated classification, synthetic data generation and developer productivity.

— A.P. Moller MAERSK *Software Engineer JL3*

10/2021 - 07/2024 *Bangalore, India*

Key Contributions:

- Designed and developed Spring Boot **WebFlux** microservices using reactive and event-driven architecture, integrating Apache **Kafka** with **Avro**, SSL/TLS, retry and Dead Letter Topic (DLT) mechanisms, and **Temporal** workflows for reliable and reentrant processing across distributed applications.
- Developed **reusable Security** and Exception Handling **libraries** to eliminate boilerplate across microservices, while maintaining high code quality.
- Ensured production-ready software through **TDD** using JUnit, **BDD** automation using Cucumber, along with Sonar remediation and **Trivy** security vulnerability fixes, deployed and managed containerized applications using Kubernetes.

— Lowe's, India *Software Engineer*

03/2020 - 10/2021 ***Bangalore, India***

Key Contributions:

- Developed reactive, non-blocking **REST APIs** using Spring **WebFlux** and integrated third-party REST services using **WebClient**, implemented **Apache Kafka** for **event-driven** audit data processing and integrated multi-bucket **Couchbase** connectivity with **Spring Boot**.
- Worked with **ForgeRock** as an **OAuth2** authorization server for **JWT**-based **authentication & authorization**, configured **SSL certificates** for ForgeRock AM and DS components, and contributed to securing and modernizing the Identity Platform.
- Deployed containerized applications to Google Cloud Platform (**GCP**) using **Spinnaker** pipelines, performed application performance testing using Apache **JMeter**, and contributed to **Sonar** remediation, **unit testing**, and **integration testing.**

AWARDS & RECOGNITIONS

Business Impact Award (12/2024)

- *Recognition of Outstanding Contribution to AWM Tech India, J.P Morgan Chase & Co.*

Hackathon Outstanding Performer (11/2025)

- *Recognized for demonstrating proactive and exceptional effort in the GraphQL Ref Data GPT AI Agent hackathon at J.P Morgan Chase & Co.*

Star Award (10/2022)

- *Recognized for exceptional performance and valuable contributions to the OHM Project at A.P. Moller - Maersk*

Valuable Player Award (12/2017)

- *Recognized as a key contributor to the Allianz Turkey Project.*

EDUCATION

Computer Science & Engineering (B. Tech-2015)

Biju Patnaik University of Technology, Rourkela, Odisha, India.